



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO K.P.D SERVICE STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

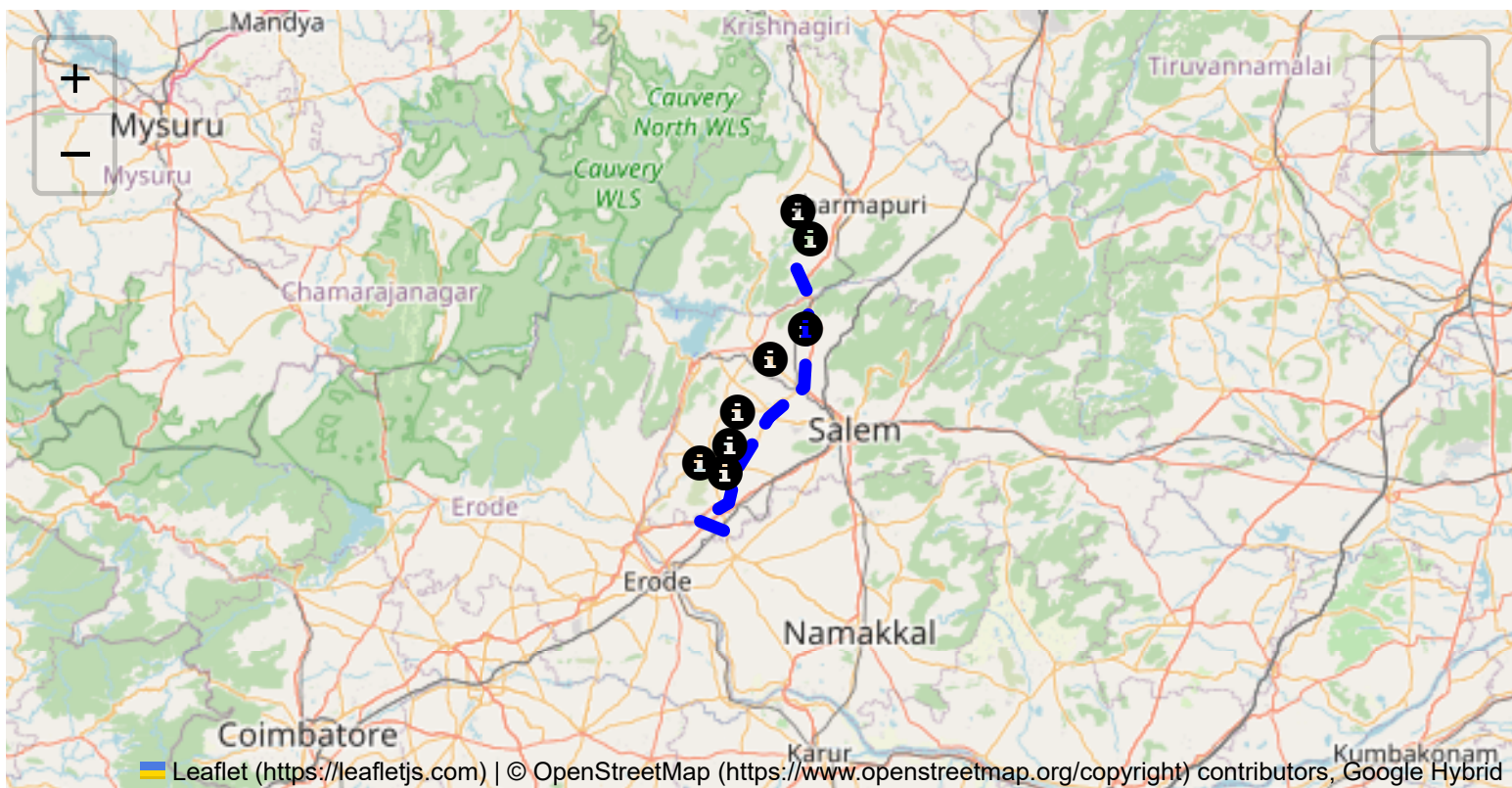
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

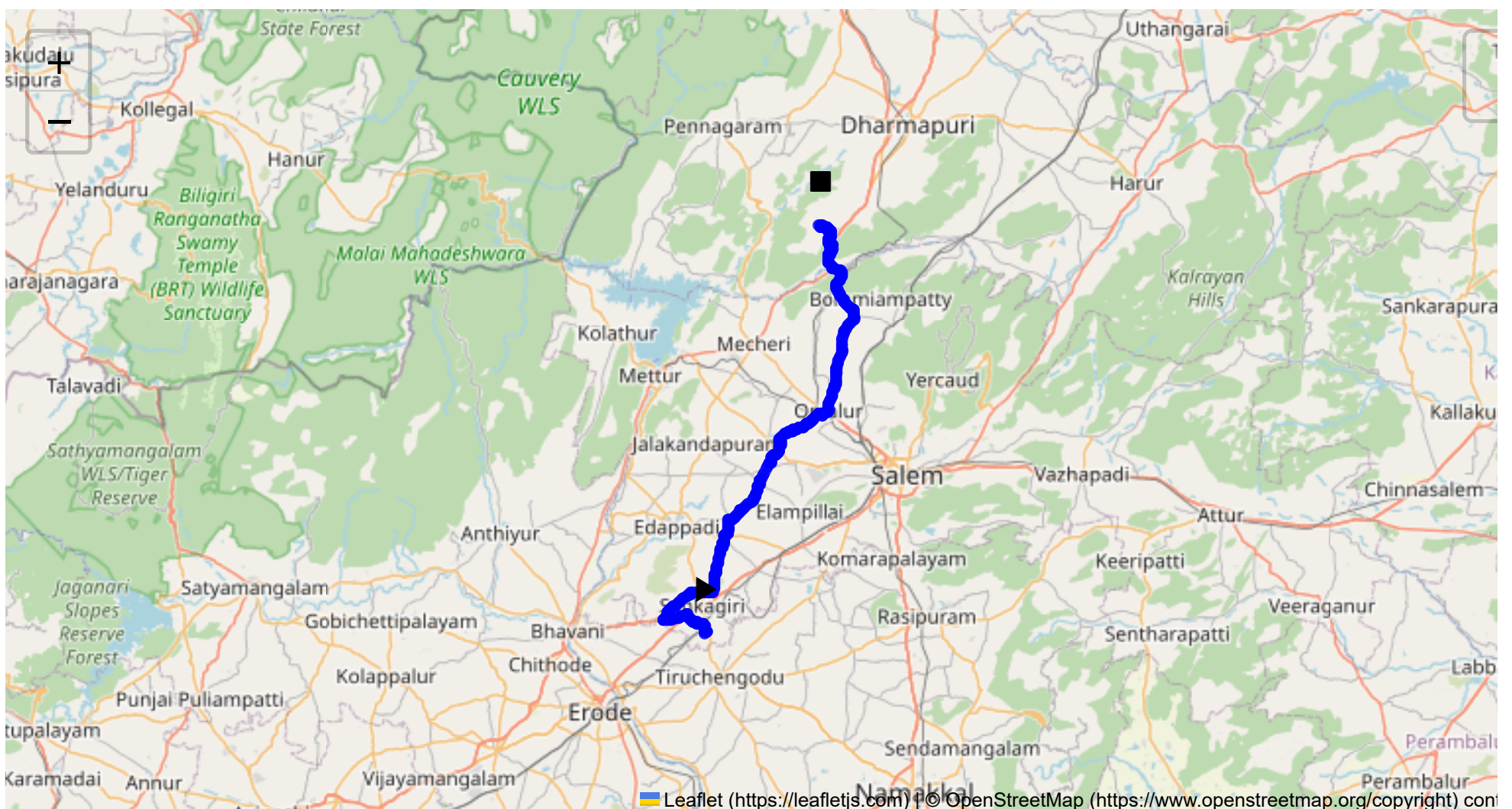
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 87.41 km
Estimated Duration: 2.0 hours
Adjusted Duration (Heavy Vehicle): 2.5 hours
Start: (11.4381, 77.8734)
End: (11.99574, 78.033083)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Maniathahalli in Tamil Nadu spans approximately 87.41 kilometers and primarily follows major state highways and local roads. This journey involves traveling through several small towns and more urban areas, which are typically connected by arterial roads. The road conditions,

traffic patterns, and environmental settings make it crucial to assess the potential hazards thoroughly for heavy vehicles transporting hazardous materials.

1. Route Map Overview

The key locations along this route include Sangagiri, Puthur, Padaiveedu, Salai - Kochi Highway, Erumaipatti, Desavilakku, Omalur, and Thoppur. From start to end, the route includes both state highways and smaller rural roads that are frequently used by local traffic, creating variable driving conditions.

2. Weather Conditions

Tamil Nadu generally experiences a tropical climate with consistent patterns of heat and humidity. From June to September, the region witnesses monsoon rains, which can lead to flooding and road slipperiness. During the summer months, high temperatures can affect engine performance and tire pressure in heavy vehicles.

3. Traffic Patterns

Traffic peaks typically occur during morning hours (8:00 AM to 10:00 AM) and evening hours (5:00 PM to 8:00 PM) due to local commuting. Congestion-prone areas include town entries and exits, especially near Omalur and Thoppur. Local festivals and market days may also increase traffic unpredictably.

4. Road Quality and Infrastructure

The roads vary from well-paved highways to narrower rural routes. Key highways such as the Salem - Kochi Highway are generally well-maintained. However, smaller roads, particularly those entering and exiting towns, may suffer from potholes and limited shoulder space.

5. Alternative Routes for Emergencies

In emergencies, detours may involve using other regional highways that parallel the planned route, such as the NH44 for faster travel between major intersections, allowing circumvention of towns if needed.

6. Regulations on Hazardous Material Transport

Local regulations in Tamil Nadu enforce strict guidelines on the transport of hazardous materials, requiring clearly marked vehicles and stringent safety measures. Night travel may be restricted in certain regions to minimize accident risks.

7. Historical Incidents

There have been reports of incidents involving heavy vehicles on the Salem - Kochi Highway due to high traffic density and treacherous road curves. Ensuring regular vehicle maintenance and compliance with load limits are crucial preventive measures.

8. Environmental and Sensitive Areas

Environmental sensitivity includes agricultural lands and proximity to residential zones, particularly around towns like Omalur and Thoppur. Noise pollution and potential chemical spills are critical concerns that should be minimized.

9. Communication Coverage

Overall, communication coverage is relatively good across the route, though potential dead zones may occur in less populated stretches, particularly in rural areas between major towns.

10. Emergency Response Times

Response times may vary, with urban areas like Salem offering quicker access to emergency services (approximately 15-30 minutes), while rural areas could take upwards of 45 minutes to an hour.

11. Overall Risk Assessment

The route poses moderate risks mainly due to variable traffic conditions, weather impacts, and areas with poor road maintenance. Adherence to safety regulations, awareness of weather conditions, and timely communication with logistic support can effectively manage these risks.

Implementing regular training sessions for drivers on emergency procedures, communication protocols, and adhering to safe driving practices is essential. Preparing for unexpected detours and ensuring the vehicle is equipped for potential mechanical issues is also advised. Additionally, maintaining compliance with all regulatory requirements on hazardous material transport will significantly mitigate risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43961, 77.87341	15 KM/Hr
1	Turn	High	11.44024, 77.87528	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	Medium	11.45357, 77.85789	30 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45847, 77.85140	30 KM/Hr
6	Turn	Medium	11.45898, 77.85135	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	Medium	11.45543, 77.81740	30 KM/Hr
9	Turn	High	11.45631, 77.81668	15 KM/Hr
10	Turn	High	11.49383, 77.88442	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
11	Turn	Medium	11.56083, 77.89921	30 KM/Hr
12	Turn	Medium	11.56180, 77.90019	30 KM/Hr
13	Turn	Medium	11.57048, 77.90224	30 KM/Hr
14	Turn	Medium	11.64217, 77.95200	30 KM/Hr
15	Turn	Medium	11.64712, 77.95118	30 KM/Hr
16	Turn	High	11.67858, 77.96834	15 KM/Hr
17	Turn	High	11.70233, 77.97869	15 KM/Hr
18	Turn	Medium	11.71394, 77.99572	30 KM/Hr
19	Turn	Medium	11.71916, 78.01117	30 KM/Hr
20	Turn	Medium	11.72559, 78.01758	30 KM/Hr
21	Turn	Medium	11.73787, 78.03528	30 KM/Hr
22	Turn	High	11.73651, 78.03868	15 KM/Hr
23	Turn	Medium	11.74120, 78.04588	30 KM/Hr
24	Turn	Medium	11.74309, 78.04725	30 KM/Hr
25	Turn	Medium	11.87998, 78.08170	30 KM/Hr
26	Turn	Medium	11.91266, 78.05852	30 KM/Hr
27	Turn	Medium	11.93735, 78.05344	30 KM/Hr
28	Turn	Medium	11.95498, 78.04689	30 KM/Hr
29	Turn	High	11.95847, 78.04989	15 KM/Hr
30	Turn	Medium	11.96011, 78.04944	30 KM/Hr
31	Turn	Medium	11.96366, 78.05142	30 KM/Hr
32	Turn	Medium	11.96427, 78.05118	30 KM/Hr
33	Turn	Medium	11.96526, 78.04950	30 KM/Hr
34	Turn	Medium	11.96580, 78.04923	30 KM/Hr
35	Turn	Medium	11.96637, 78.04923	30 KM/Hr
36	Turn	High	11.97020, 78.05123	15 KM/Hr
37	Turn	Medium	11.97313, 78.04769	30 KM/Hr
38	Turn	High	11.97400, 78.04734	15 KM/Hr
39	Turn	High	11.97525, 78.04803	15 KM/Hr
40	Turn	Medium	11.97657, 78.04775	30 KM/Hr
41	Turn	High	11.97938, 78.04918	15 KM/Hr
42	Turn	High	11.97969, 78.04903	15 KM/Hr
43	Turn	High	11.98066, 78.04958	15 KM/Hr
44	Turn	High	11.98163, 78.04890	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
45	Turn	Medium	11.98429, 78.04983	30 KM/Hr
46	Turn	Medium	11.98642, 78.05015	30 KM/Hr
47	Turn	Medium	11.98674, 78.05044	30 KM/Hr
48	Turn	High	11.98698, 78.05046	15 KM/Hr
49	Turn	High	11.98953, 78.04564	15 KM/Hr
50	Turn	Medium	11.99065, 78.04578	30 KM/Hr
51	Turn	Medium	11.99504, 78.03812	30 KM/Hr
52	Turn	Medium	11.99494, 78.03768	30 KM/Hr
53	Turn	Medium	11.99544, 78.03662	30 KM/Hr
54	Turn	Medium	11.99495, 78.03516	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
8	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
9	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
16	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
17	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
18	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
20	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44024, 77.87528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49383, 77.88442



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56083, 77.89921



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56180, 77.90019



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57048, 77.90224



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70233, 77.97869



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71394, 77.99572



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71916, 78.01117



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72559, 78.01758



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73787, 78.03528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03868



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74120, 78.04588



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74309, 78.04725



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87998, 78.08170



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91266, 78.05852



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93735, 78.05344



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95498, 78.04689



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95847, 78.04989



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

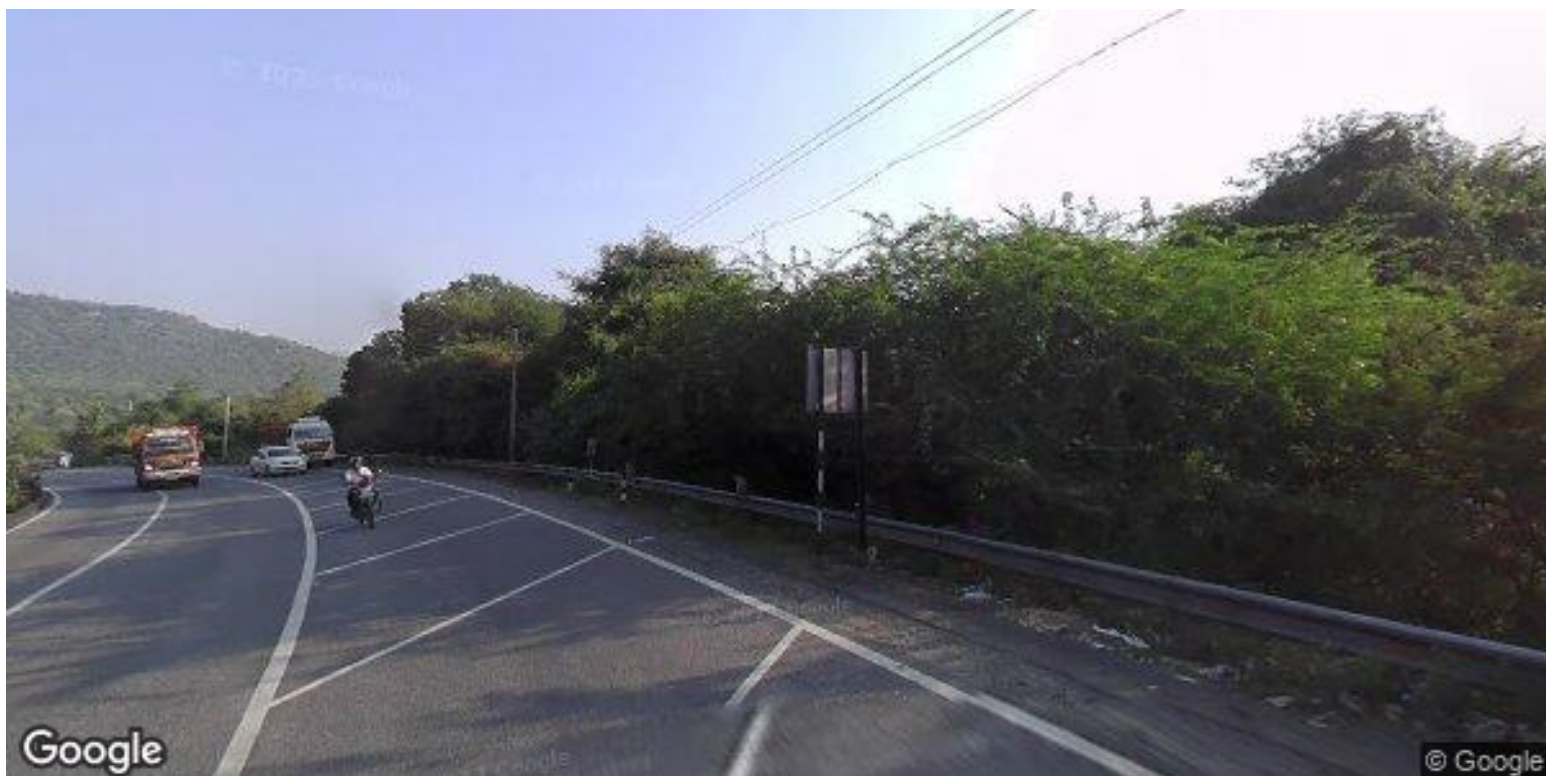
Coordinates: 11.96011, 78.04944



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96366, 78.05142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96427, 78.05118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96526, 78.04950



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96580, 78.04923



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96637, 78.04923



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.97020, 78.05123



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.97313, 78.04769



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.97400, 78.04734



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.97525, 78.04803



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.97657, 78.04775



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.97938, 78.04918



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.97969, 78.04903



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.98066, 78.04958



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.98163, 78.04890



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.98429, 78.04983



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.98642, 78.05015



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.98674, 78.05044



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.98698, 78.05046



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.98953, 78.04564



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.99065, 78.04578



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99504, 78.03812



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99494, 78.03768



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99544, 78.03662



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.99495, 78.03516
